

VANATVAM

Booking Management System

Hosting & Database Manual

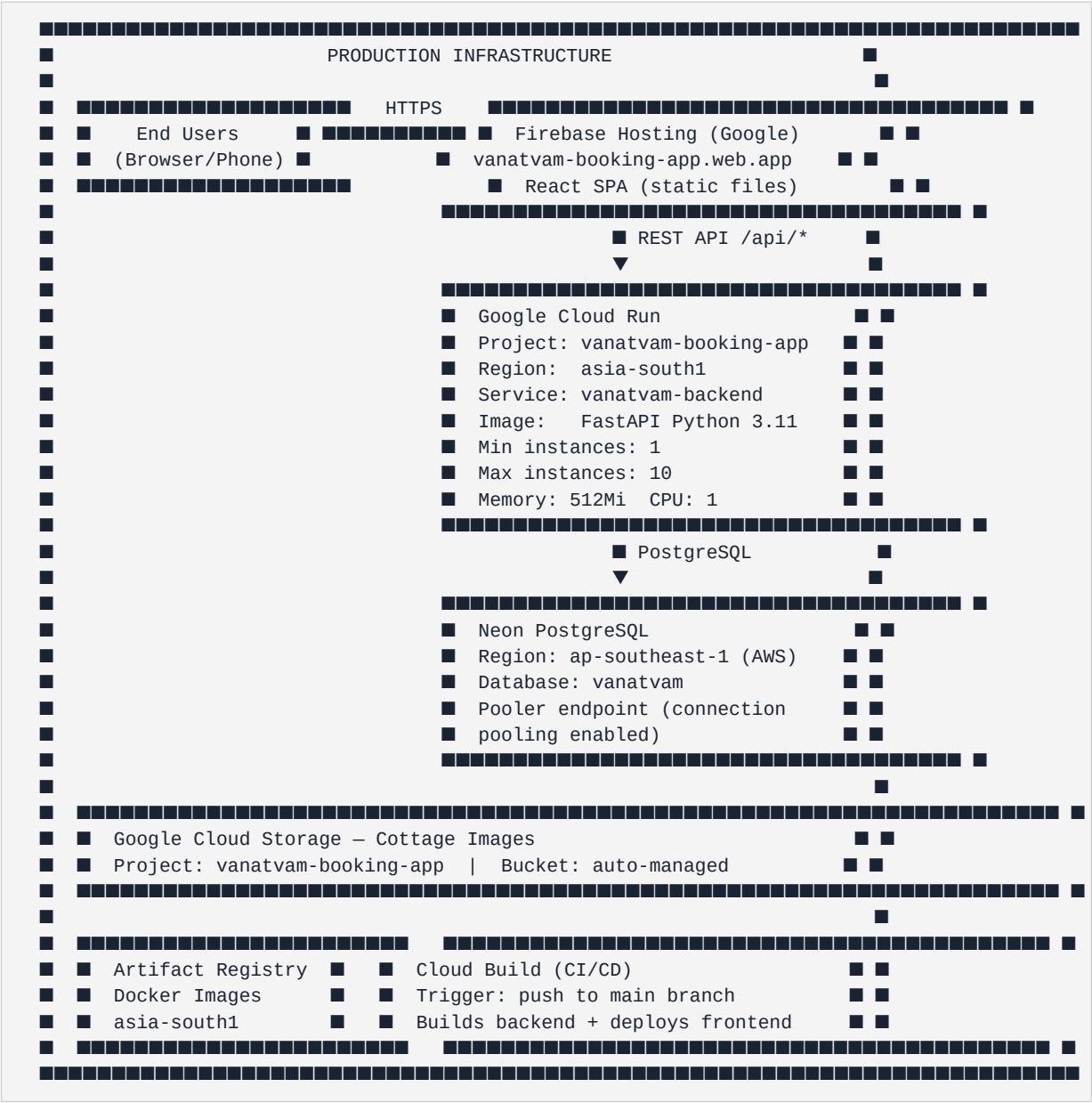
Infrastructure, Deployment & Operations Guide

Vanatvam — Hosting & Database Manual

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1. Infrastructure Overview



2. Frontend — Firebase Hosting

Service Details

Provider	Google Firebase Hosting
Project ID	`vanatvam-booking-app`
Live URL	`https://vanatvam-booking-app.web.app`
Alternate URL	`https://vanatvam-booking-app.firebaseio.com`
Plan	Spark (Free tier)
Region	Global CDN

Serve dir	`frontend/build/` (React production build)

Configuration (frontend/firebase.json)

```
{
  "hosting": {
    "public": "build",
    "rewrites": [
      { "source": "**", "destination": "/index.html" }
    ],
    "headers": [
      {
        "source": "/static/**",
        "headers": [{ "key": "Cache-Control", "value": "public, max-age=31536000, immutable" }]
      }
    ]
  }
}
```

Key settings:

- All routes rewrite to /index.html — required for React Router (SPA)
- Static assets (/static/js/, /static/css/) cached 1 year (immutable)
- Firebase serves on global CDN — fast worldwide

Deploy Command

```
# Build React app
cd frontend
npm run build

# Deploy to Firebase
npx firebase-tools deploy --only hosting --project vanatvam-booking-app
```

Firebase Free Tier Limits

Storage	10 GB
Data transfer	360 MB/day
Custom domain	Supported
SSL certificate	Automatic (free)

3. Backend — Google Cloud Run

Service Details

Provider	Google Cloud Run (Serverless containers)
GCP Project	`vanatvam-booking-app`
Project Number	`57399834436`
Region	`asia-south1` (Mumbai)
Service Name	`vanatvam-backend`
Service URL	`https://vanatvam-backend-57399834436.asia-south1.run.app`
Docker Image	`asia-south1-docker.pkg.dev/vanatvam-booking-app/vanatvam-repo/backend:latest`
Port	`8000`
Min Instances	`1` (always warm, no cold start)
Max Instances	`10`
Memory	`512Mi`
CPU	`1 vCPU`
Request Timeout	`300 seconds`
Concurrency	`80 requests/instance`
Auth	Unauthenticated (public, app-level JWT)

Runtime Environment Variables

```

DATABASE_URL=postgresql://neondb_owner:***@ep-plain-king-a1mv0lof-pooler.ap-southeast-1.aws.neon.tech/vanatvam
SECRET_KEY=<jwt-signing-key>
ALGORITHM=HS256
ACCESS_TOKEN_EXPIRE_MINUTES=30

```

■ ■ These are stored as **plain environment variables** directly on the Cloud Run service (not Secret Manager). To change them, redeploy with updated `--update-env-vars`.

Startup Probe

```

startupProbe:
  initialDelaySeconds: 30 # Wait 30s before first check
  failureThreshold: 3     # Allow 3 failures before killing
  periodSeconds: 10
  tcpSocket:
    port: 8000
  timeoutSeconds: 10

```

The 30-second initial delay gives FastAPI time to:

1. Import all modules
2. Connect to Neon PostgreSQL
3. Run `Base.metadata.create_all()` (schema sync)

4. Start uvicorn on port 8000

Dockerfile

```
FROM python:3.11-slim
WORKDIR /app
RUN apt-get update && apt-get install -y gcc libpq-dev && rm -rf /var/lib/apt/lists/*
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY . .
EXPOSE 8000
CMD ["uvicorn", "main:app", "--host", "0.0.0.0", "--port", "8000"]
```

CORS Allowed Origins

```
allow_origins = [
    "http://localhost:3000",          # Local development
    "https://vanatvam-booking.web.app", # Firebase (legacy)
    "https://vanatvam-booking.firebaseio.com", # Firebase (legacy alt)
    "https://vanatvam-booking-app.web.app", # Firebase (current)
    "https://vanatvam-booking-app.firebaseio.com",
]
```

Redeploy Backend Command

```
# Build and push image (requires Docker Desktop running)
docker build -t asia-south1-docker.pkg.dev/vanatvam-booking-app/vanatvam-repo/backend:latest ./backend
docker push asia-south1-docker.pkg.dev/vanatvam-booking-app/vanatvam-repo/backend:latest

# OR build using Cloud Build (no local Docker needed)
gcloud builds submit ./backend \
  --tag asia-south1-docker.pkg.dev/vanatvam-booking-app/vanatvam-repo/backend:latest \
  --project=vanatvam-booking-app \
  --region=asia-south1

# Deploy to Cloud Run
gcloud run deploy vanatvam-backend \
  --image=asia-south1-docker.pkg.dev/vanatvam-booking-app/vanatvam-repo/backend:latest \
  --region=asia-south1 \
  --project=vanatvam-booking-app
```

Update Environment Variables

```
gcloud run services update vanatvam-backend \
  --region=asia-south1 \
  --project=vanatvam-booking-app \
  --update-env-vars="KEY=VALUE"
```

4. Database — Neon PostgreSQL

Service Details

Provider	Neon (neon.tech) — Serverless PostgreSQL
Host Region	AWS ap-southeast-1 (Singapore)
Database Name	`vanatvam`
Username	`neondb_owner`
Connection Type	Pooler (PgBouncer) — optimised for serverless
Pooler Host	`ep-plain-king-a1mv0lof-pooler.ap-southeast-1.aws.neon.tech`
SSL	Required (`sslmode=require`)
Plan	Free tier

Full Connection String

```
postgresql://neondb_owner:<password>@ep-plain-king-a1mv0lof-pooler.ap-southeast-1.aws.neon.tech/vanatvam?sslmode=require
```

■ Store the actual password securely. Do not commit it to the repository.

Why Neon?

- **Serverless:** Database scales to zero when idle (cost-efficient)
- **Connection pooling:** Built-in PgBouncer handles many concurrent connections from Cloud Run instances
- **Branching:** Supports database branches for testing (Git-like)
- **Free tier:** Sufficient for current load

Database Schema (Tables)

`users`	All system users (admins and owners)
`properties`	Sanctuaries/properties grouping cottages
`cottages`	Individual bookable cottage units
`bookings`	All booking records with status
`maintenance_blocks`	Cottage unavailability periods for maintenance
`system_calendar`	Holiday and peak season date flags
`peak_seasons`	Named peak season date ranges
`quota_transactions`	Full audit log of all credit movements
`email_configs`	SMTP configuration for transactional email
`email_templates`	HTML/text templates for each email type

SQLAlchemy Connection Pool Settings (backend/database.py)

```
engine = create_engine(
    DATABASE_URL,
    pool_size=5,           # Maintain 5 persistent connections
    max_overflow=10,       # Allow up to 10 additional connections
    pool_recycle=300,      # Recycle connections every 5 minutes
    pool_pre_ping=True,    # Test connection health before use
)
```

Schema Management

The backend auto-syncs the schema on startup:

```
Base.metadata.create_all(bind=engine)
```

This creates any missing tables but does NOT drop existing ones. For schema changes (new columns), use migration scripts (add_email_columns.py, add_email_tables.py).

Backup Strategy

Neon Free tier provides:

- Point-in-time restore (limited window on free tier)
- For additional backup, run periodic pg_dump:

```
pg_dump "postgresql://neondb_owner:<password>@ep-plain-king-a1mv0lof-pooler.ap-southeast-1.aws.neon.tech/van"
```

Accessing the Database (Admin Console)

1. Go to console.neon.tech
2. Select the vanatvam project
3. Use the SQL editor for direct queries
4. Or connect with psql:

```
psql "postgresql://neondb_owner:<password>@ep-plain-king-a1mv0lof-pooler.ap-southeast-1.aws.neon.tech/van"
```

5. Image Storage — Google Cloud Storage

Provider	Google Cloud Storage
Project	`vanatvam-booking-app`
Access	Managed via Cloud Run service account
Usage	Cottage images uploaded by admin
Upload API	`POST /api/admin/cottages/{id}/upload-image`

Images are uploaded from the admin UI → FastAPI backend → GCS. The resulting public URL is stored in the cottages.image_url column.

6. Environment Variables

Local Development (backend/.env)

```
DATABASE_URL=postgresql://neondb_owner:<password>@ep-plain-king-a1mv0lof-pooler.ap-southeast-1.aws.neon.tech
SECRET_KEY=<your-jwt-secret>
ALGORITHM=HS256
ACCESS_TOKEN_EXPIRE_MINUTES=30
```

Frontend Production (frontend/.env.production)

```
REACT_APP_API_URL=https://vanatvam-backend-57399834436.asia-south1.run.app
```

Frontend Local Development (frontend/.env — create if missing)

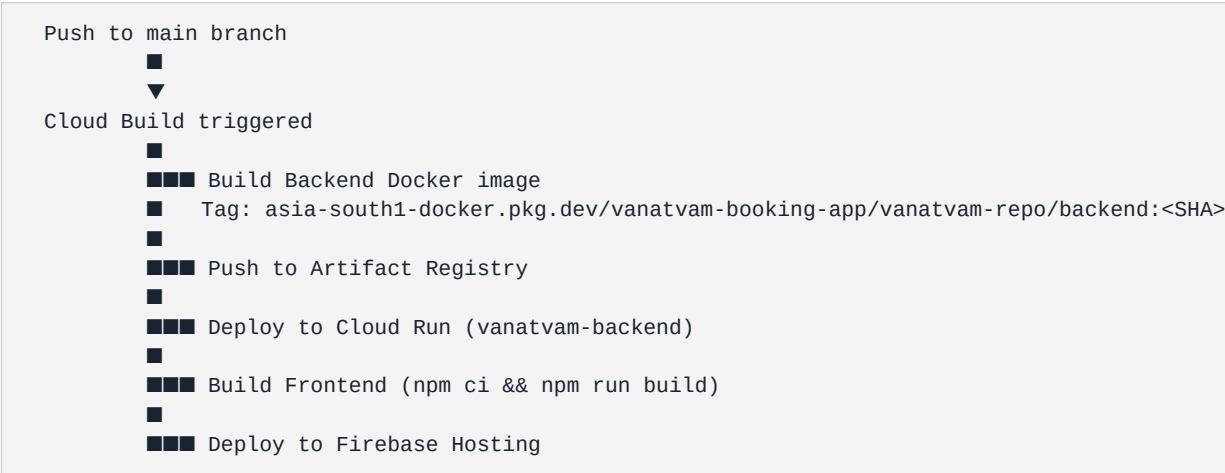
```
REACT_APP_API_URL=http://localhost:8000
```

Cloud Run Environment (set via gcloud)

All backend/.env variables are set directly on the Cloud Run service as environment variables. There is no Secret Manager in use.

7. CI/CD — Cloud Build

Pipeline (cloudbuild.yaml)



Artifact Registry

Location	`asia-south1`

Repository	`vanatvam-repo`
Format	Docker
Full path	`asia-south1-docker.pkg.dev/vanatvam-booking-app/vanatvam-repo/backend`

8. Deployment Procedures

Deploy Frontend Only

```
cd frontend
npm run build
npx firebase-tools deploy --only hosting --project vanatvam-booking-app
```

Deploy Backend Only (no local Docker)

```
# Build on Cloud Build (uploads code, builds in cloud)
gcloud builds submit ./backend \
  --tag asia-south1-docker.pkg.dev/vanatvam-booking-app/vanatvam-repo/backend:latest \
  --project=vanatvam-booking-app \
  --region=asia-south1

# Deploy new revision
gcloud run deploy vanatvam-backend \
  --image=asia-south1-docker.pkg.dev/vanatvam-booking-app/vanatvam-repo/backend:latest \
  --region=asia-south1 \
  --project=vanatvam-booking-app
```

Update Backend Config (env vars only, no rebuild)

```
gcloud run services update vanatvam-backend \
  --region=asia-south1 \
  --project=vanatvam-booking-app \
  --update-env-vars="DATABASE_URL=<new-url>,SECRET_KEY=<new-key>"
```

Reset Admin Password

```
cd backend
py reset_password.py
# Edit the script first to set email and new password
```

Or run against the Neon DB directly:

```

from database import SessionLocal
from models import User
from auth import get_password_hash

db = SessionLocal()
user = db.query(User).filter(User.email == "admin@vanatvam.com").first()
user.password_hash = get_password_hash("newpassword")
db.commit()
db.close()

```

9. Monitoring & Logs

Cloud Run Logs

```

# Stream live logs
gcloud run services logs tail vanatvam-backend \
  --region=asia-south1 \
  --project=vanatvam-booking-app

# Read last 50 log entries
gcloud run services logs read vanatvam-backend \
  --region=asia-south1 \
  --project=vanatvam-booking-app \
  --limit=50

```

GCP Console Links

Cloud Run	https://console.cloud.google.com/run?project=vanatvam-booking-app
Artifact Registry	https://console.cloud.google.com/artifacts?project=vanatvam-booking-app
Cloud Build	https://console.cloud.google.com/cloud-build/builds?project=vanatvam-booking-app
Firebase Console	https://console.firebase.google.com/project/vanatvam-booking-app

Health Check

```

curl https://vanatvam-backend-57399834436.asia-south1.run.app/
# Expected: {"message": "Vanatvam API is running"}

```

API Docs (FastAPI Swagger UI)

```
https://vanatvam-backend-57399834436.asia-south1.run.app/docs
```

10. Cost Summary

Firebase Hosting	Spark (Free)	**Free**
Cloud Run	Min 1 instance, ~512Mi	~\$5–8/month
Artifact Registry	<0.5GB storage	**Free**
Cloud Build	<120 min/day	**Free**
Neon PostgreSQL	Free tier	**Free**
Cloud Storage (images)	Small volume	~\$0–1/month
Total		**~\$5–9/month**

■■ **Billing must be enabled** on GCP project `vanatvam-booking-app` for Cloud Run to function. Firebase Hosting and Neon remain free regardless.

11. Troubleshooting

Backend returns 429 "Rate exceeded"

Cause: Cloud Run startup probe fires before FastAPI finishes initializing.

Fix: Already resolved — startup probe has `initialDelaySeconds: 30`.

If it recurs after a redeploy:

```
gcloud run services replace cloudrun-service.yaml --region=asia-south1 --project=vanatvam-booking-app
```

CORS error in browser

Symptom: No 'Access-Control-Allow-Origin' header

Check:

1. Is the backend URL correct in frontend/.env.production?
2. Is the frontend URL in backend/main.py `allow_origins` list?
3. Is the backend running? Test: `curl /`

Backend 500 on startup

Most likely causes:

1. DATABASE_URL env var missing or wrong → check Cloud Run env vars
2. Neon database is paused (free tier auto-pauses) → visit Neon console to wake it

```
# Check current env vars on Cloud Run
gcloud run services describe vanatvam-backend \
  --region=asia-south1 \
  --project=vanatvam-booking-app \
  --format="yaml(spec.template.spec.containers[0].env)"
```

Database connection refused

Symptom: connection to server at "localhost" refused

Cause: DATABASE_URL env var not set on Cloud Run — fallback to localhost:5432 being used.

Fix: Set the env var:

```
gcloud run services update vanatvam-backend \
  --region=asia-south1 \
  --project=vanatvam-booking-app \
  --update-env-vars="DATABASE_URL=postgresql://neondb_owner:<password>@ep-plain-king-a1mv0lof-pooler.ap-sou
```

Firebase deploy fails: "No currently active project"

```
npx firebase-tools deploy --only hosting --project vanatvam-booking-app
# Always pass --project explicitly
```

Cannot login to admin

Admin password can be reset by running `reset_password.py` locally (connects to Neon DB using `backend/.env`):

```
cd backend
py reset_password.py
```

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